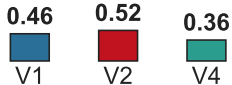


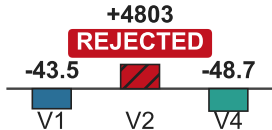
# Physics-gated AI miniprotein design on VEGF-A

vdW gate beats ipTM · 1FLT template beats MSA

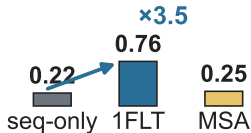
## 1 · ipTM picks V2



## 2 · vdW gate (kcal/mol)



## 3 · 1FLT vs MSA



**vdW-gated ranking:  $V1 \geq V4 > V2$**

V1 leads in every predictor; V2 fails the gate in every mode.

V4 bridges the VEGF-A dimer (ipTM 0.66–0.68); V1 does not.